What Lean Cannot Say: A Machine-Checked Analysis of Wittgenstein's *Tractatus*

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Abstract

We present a machine-checked reconstruction of the semantic core of *Tractatus Logico-Philosophicus* in the Lean proof assistant. The formalization models objects, states of affairs, worlds, and propositions, and proves truth-functional compositionality, functional completeness, and bivalence.

Beyond encoding, the system is parameterized over world models, allowing a precise separation between (i) structural invariants of the propositional syntax, (ii) model-dependent assumptions such as the independence of atomic facts, and (iii) formal limits of expressibility. We show that truth-functional compositionality holds in all models, while independence characterizes a specific "free" model and fails in constrained models.

To analyze the relation between object language and metalanguage, we introduce an `expresses` relation linking propositions to meta-level predicates. Using this, we prove that any proposition expressing a world-independent property must be a tautology or contradiction. Consequently, nontrivial propositions cannot express world-independent truths.

We further distinguish three notions of equivalence—syntactic identity, logical-form equivalence up to permutation, and semantic equivalence—and prove their strict separation.

These results yield a formal decomposition of the *Tractatus* into invariants, assumptions, and limits, and provide a precise characterization of the saying/showing distinction as a theorem about expressibility in truth-functional languages.

1 Introduction

Wittgenstein's *Tractatus Logico-Philosophicus* [1] proposes that the world consists of atomic facts (Sachverhalte), that propositions are truth-functions of elementary propositions, and that the limits of language are the limits of the world. The text is terse, aphoristic, and deliberately resistant to formalization—yet its semantic core is a precise theory of truth-functional composition that invites formal reconstruction.

Several authors have pursued this invitation. Lokhorst [2] provides a set-theoretic reconstruction covering ontology, semantics, and propositional attitudes; Miller [7] develops a Tractarian semantics for predicate logic; Weiss [4] reconstructs the logical system with Π^1_1-completeness results; Stokhof [3] traces the *Tractatus*'s influence on formal semantics. These reconstructions clarify the *Tractatus* as a semantic system but are carried out on paper, without machine verification. Separately, the proof-assistant community has formalized philosophical arguments [8, 9] and even complete metaphysical systems [10], but a Tractarian formalization has not appeared.

This paper contributes a machine-checked reconstruction in Lean 4 [12] that goes beyond encoding to yield new analytical results. Our central claim is:

> *The saying/showing distinction of the* Tractatus *can be reformulated as a theorem: nontrivial propositions cannot express world-independent truths.*

This is not a paraphrase of Wittgenstein but a provable consequence of the typed structure of the formalization. The key device is an `expresses` relation (Definition 5.1) that bridges the object language (propositions of type Proposition S) and the metalanguage (terms of type Prop), making the object/meta boundary visible at the type level.

More broadly, by parameterizing the formalization over a general `WorldModel` structure, we show that the claims of the *Tractatus* decompose into three classes:

1. **Structural invariants** that hold in every world model—compositionality, bivalence, De Morgan duality.
2. **Model-dependent assumptions** that hold in the unconstrained "free" model but fail in constrained models—principally, the independence of atomic facts.
3. **Formal limits** on expressibility—the collapse of world-independent content to tautology or contradiction, and the strict separation of syntactic, logical-form, and semantic equivalence.

The formalization comprises 1,517 lines of Lean 4, proves 46 theorems with 0 `sorry` obligations and 1 deliberate axiom (`axiom silence : True`, encoding TLP 7), and is publicly available.[1]

The paper is organized as follows. Section 2 presents the formalization: types, evaluation, and the world-model abstraction. Section 3 establishes the structural invariants. Section 4 analyzes the model-dependent assumptions and exhibits countermodels. Section 5 proves the expressibility results and the equivalence hierarchy. Section 6 discusses related work. Section 7 reflects on the philosophical implications and limitations.

2 The Formalization

We formalize the ontological skeleton of the *Tractatus* in Lean 4. The design choices are deliberate modeling decisions, not exegetical claims about Wittgenstein's intent. We state them explicitly so that the reader can assess the reconstruction's scope and limitations.

2.1 Objects and States of Affairs

TLP 2.02: "The object is simple." Objects are the substance of the world (TLP 2.021). We model them as an abstract type `TractObject` with no internal structure, capturing Wittgenstein's insistence on simplicity.

A state of affairs (Sachverhalt) is a possible combination of objects (TLP 2.01). Rather than encoding the combinatorial structure of objects into states of affairs, we index states of affairs by an abstract type S:

```
variable (TractObject : Type)
variable (Sachverhalt : Type)
```

This choice leaves the internal structure of states of affairs unspecified—an interpretive decision that captures independence while remaining agnostic about combinatorial form (cf. TLP 2.0141).

2.2 Worlds

TLP 1: "The world is everything that is the case." A world is determined by which states of affairs obtain. We model this as a predicate:

```
def World := Sachverhalt →Prop
```

Every function $\mathsf{S} \to \mathsf{Prop}$ counts as a world. This gives the full Boolean cube of truth-value assignments, making the independence of atomic facts (TLP 2.061) hold by construction. We return to this modeling choice and its consequences in Section 4.

2.3 Propositions

TLP 5: "The proposition is a truth-function of elementary propositions." We define propositions as an inductive type built from elementary propositions, negation, and conjunction:

```
inductive Proposition (S : Type) where
  | elementary : S →Proposition S
```

[1] https://leangenius.org/proof/tractatus-ontology

```
  | neg       : Proposition S →Proposition S
  | conj      : Proposition S →Proposition S →Proposition S
```

Disjunction, implication, biconditional, and NAND are defined as derived operations (TLP 5.101, 5.5), confirming that $\{\neg, \wedge\}$ is an adequate basis.

2.4 Semantic Evaluation

The core semantic function interprets propositions against a world:

```
def Proposition.eval (p : Proposition S) (w : World S) : Prop :=
  match p with
  | .elementary s => w s
  | .neg q       => ¬(q.eval w)
  | .conj q r    => q.eval w ∧r.eval w
```

An elementary proposition `elementary` s is true in world w exactly when $w\ s$ holds. Negation and conjunction lift to their classical counterparts. We also provide a `Bool`-valued evaluator `evalBool` for decidable computation and prove its correctness against `eval`.[2]

2.5 Tautology, Contradiction, Nontriviality

A proposition is a *tautology* if it holds in every world, a *contradiction* if it holds in none, and *nontrivial* if it holds in some world and fails in another:

```
def IsTautology (p : Proposition S) : Prop :=
  ∀ w : World S, p.eval w
def IsContradiction (p : Proposition S) : Prop :=
  ∀ w : World S, ¬(p.eval w)
def Nontrivial (p : Proposition S) : Prop :=
  ∃ w₁ w₂ : World S, p.eval w₁ ∧¬p.eval w₂
```

We prove that nontriviality is equivalent to being neither a tautology nor a contradiction (Theorem `nontrivial_iff_not_taut`

2.6 General World Models

The definitions above hardwire the world model: every function $\mathsf{S} \to \mathsf{Prop}$ is a world. In many applications—physics, epistemic logic, constrained databases—not every assignment is legitimate. We introduce a `WorldModel` structure to parameterize the semantics:

```
structure WorldModel (S : Type) where
  W       : Type
  holds   : W →S →Prop
  nonempty : Nonempty W
```

The *free model* `freeModel` recovers the original semantics: $W = \mathsf{S} \to \mathsf{Prop}$ and `holds` is function application. We define a general evaluator `evalM` and prove that `evalM (freeModel S)` agrees with `Proposition.eval` (Theorem `evalM_free_eq_eval`).

This abstraction is the key structural device of the formalization. It allows us to state precisely which results depend on the choice of world model and which do not.

3 Structural Invariants

The following results hold in *every* world model—they are consequences of the inductive structure of propositions and classical logic alone, with no assumptions about which worlds exist.

[2] Theorem `evalBool_correct`: `(p.evalBool w = true) ↔ p.eval (fun s ⇒ w s = true)`.

3.1 Truth-Functional Compositionality

Theorem 3.1 (Compositionality, model-independent)**.** *For any world model M, any proposition p, and any two worlds $w_1, w_2 \in M.W$: if $M.holds\ w_1\ s \leftrightarrow M.holds\ w_2\ s$ for every state of affairs s, then* $\mathsf{eval}_M\ p\ w_1 \leftrightarrow \mathsf{eval}_M\ p\ w_2$.

The proof is by structural induction on propositions. The elementary case follows from the hypothesis; negation and conjunction lift via `Iff.not` and `Iff.and`.

```
theorem truth_functional_compositionality_gen (p : Proposition S)
    (w₁ w₂ : M.W)
    (h : ∀s : S, M.holds w₁ s ↔M.holds w₂ s) :
    evalM M p w₁ ↔evalM M p w₂ := by
  induction p with
  | elementary s => exact h s
  | neg q ih    => simp only [evalM]; exact ih.not
  | conj q r ihq ihr => simp only [evalM]; exact ihq.and ihr
```

This establishes TLP 5 as a structural invariant: the truth-value of a compound proposition is determined by its elementary constituents regardless of which worlds the model admits.

3.2 Bivalence and Classical Tautologies

Elementary bivalence (TLP 4.023) is an instance of classical excluded middle:

Theorem 3.2 (Bivalence)**.** *For any state of affairs s and world w: $w\ s \vee \neg\ w\ s$.*

The standard tautologies follow: excluded middle ($p \vee \neg p$), double negation ($\neg\neg p \leftrightarrow p$), De Morgan's laws, and modus ponens. We verify each as a formal theorem.

3.3 NAND Functional Completeness

TLP 5.5 anticipates the Sheffer stroke. We prove that negation and conjunction are both expressible via NAND:

Theorem 3.3 (NAND completeness)**.** *For any propositions p, q and world w:*

1. $\mathsf{nand}(p, p).\mathsf{eval}\ w \leftrightarrow (\neg p).\mathsf{eval}\ w$
2. $(\neg\ \mathsf{nand}(p, q)).\mathsf{eval}\ w \leftrightarrow (p \wedge q).\mathsf{eval}\ w$

Since $\{\neg, \wedge\}$ generates all truth-functions and NAND generates $\{\neg, \wedge\}$, NAND alone is functionally complete—confirming TLP 5.5.

4 Model-Dependent Assumptions

Not every claim in the *Tractatus* is a structural invariant. The independence of atomic facts—perhaps the most famous ontological thesis of the text—is a property of a specific world model.

4.1 Independence in the Free Model

We formalize independence as a typeclass:

```
class IndependentWorlds (S : Type) where
  realizable : ∀assignment : S →Prop,
    ∃ w : World S, ∀s, w s ↔assignment s
```

The free model trivially satisfies this: every assignment *is* a world.

```
instance : IndependentWorlds S :=
  ⟨fun a => ⟨a, fun _ => Iff.rfl⟩⟩
```

Independence is thus not a theorem about logic but a consequence of the modeling choice $\mathsf{World\ S} = \mathsf{S} \to \mathsf{Prop}$. This distinction—between what is proved and what is assumed by the encoding—is precisely the kind of clarification that a typed formalization forces.

4.2 Failure in Constrained Models

To show that independence is genuinely contingent on the model, we exhibit two countermodels.

Abstract countermodel. Consider two states of affairs a, b with $a \neq b$, where b must co-occur with a. Define `ConstrainedWorld` $\mathsf{S}\ a\ b = \{w : \mathsf{S} \to \mathsf{Prop} \mid w\ a \to w\ b\}$. Then:

Theorem 4.1 (Independence fails in constrained models)**.** *If $a \neq b$, then there is no constrained world realizing the assignment $\{a \mapsto \top, b \mapsto \bot\}$.*

```
theorem constrained_independence_fails (a b : S) (hab : a  b) :
    ¬ ∀ assignment : S →Prop,
      ∃ cw : ConstrainedWorld S a b,
        ∀ s, cw.val s ↔assignment s := by
  intro h
  let bad : S →Prop := fun s => s = a
  obtain ⟨⟨w, hw⟩, hmatch⟩ := h bad
  have ha : w a := (hmatch a).mpr rfl
  have hb : w b := hw ha
  exact hab ((hmatch b).mp hb)
```

Weather model. A more concrete example: three states of affairs (rain, clouds, snow) with the physical constraint *rain implies clouds*. The weather model restricts worlds to those satisfying this constraint.

Theorem 4.2 (Weather model)**.** *In the weather model, there is no world where rain holds but clouds does not.*

These results make explicit a point that is often asserted informally: TLP 2.061 ("Atomic facts are independent of one another") is not a logical law but a constraint on the space of possible worlds. Our formalization turns this observation into a proof.

5 Formal Limits: Expressibility and Equivalence

The deepest results of the formalization concern the boundaries of what the object language can express. These are not claims within the Tractarian system but *about* it—theorems at the meta-level that formalize Wittgenstein's saying/showing distinction.

5.1 The `expresses` Relation

We introduce a typed bridge between the object language and the metalanguage:

Definition 5.1 (`expresses`)**.** A proposition $p : \mathsf{Proposition\ S}$ *expresses* a meta-level property $P : \mathsf{Prop}$ if and only if p's truth value tracks P in every world:

$$\mathsf{expresses}\ p\ P \quad \stackrel{\text{def}}{=} \quad \forall\, w : \mathsf{World\ S},\ \ p.\mathsf{eval}\ w \leftrightarrow P$$

```
def expresses (p : Proposition S) (P : Prop) : Prop :=
  ∀ w : World S, p.eval w ↔P
```

The type signature makes the object/meta distinction visible: p lives in the object language, P lives in Lean's metalanguage, and `expresses` is the claim that they correspond. This is the formalization's key analytical device.

5.2 Expressibility Collapse

Theorem 5.2 (Expressibility collapse)**.** *If a proposition p expresses a world-independent property $P : \mathsf{Prop}$, then p is a tautology or a contradiction.*

Proof. Suppose $\mathsf{expresses}\ p\ P$ holds. Then for any two worlds w_1, w_2:

$$p.\mathsf{eval}\ w_1 \ \leftrightarrow\ P \ \leftrightarrow\ p.\mathsf{eval}\ w_2$$

so p has the same truth value in all worlds. By excluded middle on $p.\mathsf{eval}\ w_0$ for an arbitrary world w_0, either p holds everywhere (tautology) or fails everywhere (contradiction). □

```
theorem saying_showing_triviality (q : Proposition S)
    (P : Prop) [Nonempty S]
    (h : expresses q P) :
    IsTautology q ∨IsContradiction q := by
  apply world_constant_taut_or_contra
  intro w₁ w₂
  exact (h w₁).trans (h w₂).symm
```

The contrapositive is immediate and philosophically sharper:

Corollary 5.3 (Nontrivial propositions cannot express world-independent content)**.** *If p is nontrivial (i.e., true in some world and false in another), then there is no $P : \mathsf{Prop}$ such that $\mathsf{expresses}\ p\ P$.*

```
theorem nontrivial_cannot_express_world_independent
    (p : Proposition S) (P : Prop) [Nonempty S]
    (hnt : Nontrivial p) :
    ¬ expresses p P := by
  intro h
  exact nontrivial_expressibility_requires_world_dependence
    p P hnt h
```

This is the paper's headline result. It says: if a proposition genuinely "says" something—i.e., carves out a nontrivial region of logical space—then what it says cannot be a world-independent truth. Genuine content requires variation across possible worlds. What is world-independent can only be "shown" by the structure of the formalism itself, not "said" within it.

This transforms the saying/showing distinction from a philosophical thesis into a provable constraint on truth-functional languages.

5.3 The Equivalence Hierarchy

The formalization distinguishes three notions of proposition equivalence, ordered by strictness:

Definition 5.4 (Equivalence hierarchy)**.** Let $p, q : \mathsf{Proposition\ S}$.

1. **Structural equivalence** ($p \sim_{\text{struct}} q$): p and q are built from identical constructors in the same tree shape with identical atoms.
2. **Logical-form equivalence** ($p \sim_{\text{form}} q$): there exists a bijective renaming of atoms (a permutation $\sigma : \mathsf{S} \xrightarrow{\sim} \mathsf{S}$) such that $p.\mathtt{rename}\ \sigma = q$.

3. **Semantic equivalence** ($p \sim_{\text{sem}} q$): p and q have the same truth value in every world: $\forall w,\ p.\mathsf{eval}\ w \leftrightarrow q.\mathsf{eval}\ w$.

We prove that $\sim_{\text{form}}$ is an equivalence relation (reflexive by the identity permutation, symmetric by inverse, transitive by composition) and establish the strict hierarchy:

Theorem 5.5 (Strict separation)**.** $\sim_{\text{struct}} \subsetneq \sim_{\text{form}} \subsetneq \sim_{\text{sem}}$

Proof. Each inclusion is witnessed by containment and a separating example.

$\sim_{\text{struct}} \subsetneq \sim_{\text{form}}$*:* Structural equality implies propositional equality (by induction), which implies logical-form equivalence via the identity permutation. Strictness: `elementary`(*rain*) and `elementary`(*snow*) are $\sim_{\text{form}}$ (via the swap permutation) but not $\sim_{\text{struct}}$ (different atoms).

$\sim_{\text{form}} \subsetneq \sim_{\text{sem}}$*:* Logical-form equivalence implies truth-table isomorphism (Theorem `formEq_implies_truth_table_iso`), and truth-table isomorphism with trivial relabeling gives semantic equivalence. Strictness: $\neg\neg$(`elementary` s) is $\sim_{\text{sem}}$ to `elementary` s (by double negation) but not $\sim_{\text{form}}$ (renaming preserves tree depth, so a depth-3 tree cannot become depth-1). □

This hierarchy clarifies a long-standing ambiguity in the *Tractatus*. Wittgenstein's notion of "logical form" (TLP 4.0141) is neither syntactic identity nor truth-conditional equivalence but something in between. Our $\sim_{\text{form}}$—defined via bijective atom renaming—provides a formal candidate: it captures the connective structure while abstracting over which particular atoms appear. The strict separation proves that this intermediate notion is genuine, not collapsible into either extreme.

5.4 First-Order Extension

In a companion file, we extend the propositional calculus with first-order quantifiers using higher-order abstract syntax (HOAS):

```
inductive FOProp (S : Type) (D : Type) : Type where
  | lift    : Proposition S →FOProp S D
  | negFO   : FOProp S D →FOProp S D
  | conjFO  : FOProp S D →FOProp S D →FOProp S D
  | forall_ : (D →FOProp S D) →FOProp S D
  | exists_ : (D →FOProp S D) →FOProp S D
```

The HOAS encoding leverages Lean's metalanguage for variable binding, avoiding explicit substitution machinery. We prove that compositionality lifts to the first-order level:

Theorem 5.6 (First-order compositionality)**.** *For any* $p : \mathsf{FOProp}\ \mathsf{S}\ D$ *and worlds* w_1, w_2 *agreeing on all states of affairs,* $p.\mathsf{eval}\ w_1 \leftrightarrow p.\mathsf{eval}\ w_2$.

We also establish universal–existential duality (TLP 5.52), quantifier distribution, vacuous quantifier elimination, and De Morgan duality for quantifiers. The file notes the Geach–Soames critique [5, 6]: for infinite domains, Wittgenstein's claim that quantifiers reduce to applications of the N-operator is problematic, since the N-operator is a finitary syntactic operation.

6 Related Work

Formal reconstructions of the *Tractatus*. Lokhorst [2] provides the most comprehensive prior formal reconstruction, covering ontology, semantics, and propositional attitudes in a set-theoretic framework. His axiomatization of independence and his truth-functionality theorem (every sentence is a truth-function of elementary sentences) overlap with our encoding, but he does not formalize the saying/showing boundary itself. Weiss [4] gives a 50-page reconstruction of the *Tractatus*'s logical system, analyzing the N-operator and form-series in detail and proving that tautology-hood is Π^1_1-complete for countable domains—a striking complexity result that sharpens the Geach–Soames critique. Miller [7] develops a Tractarian semantics for predicate logic that is independent of the metaphysics of logical atomism. Stokhof [3] reads the *Tractatus* as a proto-formal semantics, tracing its influence through Carnap's state descriptions to possible-worlds

semantics, and argues that the saying/showing distinction arises from the universalism of the framework—a philosophical diagnosis our expressibility collapse theorem makes precise. All of these works provide formal clarity but are carried out on paper; none offers machine-checked proofs, and none parameterizes over world models to isolate which claims are structural invariants versus modeling assumptions.

The Geach–Soames critique. Geach [5] argues that the N-operator cannot ground quantification over infinite domains, since it is a finitary operation. Soames [6] sharpens this: the *Tractatus*'s claim that all of logic reduces to truth-functional operations on elementary propositions is undermined when the domain of quantification is infinite. Our formalization takes this critique seriously: the first-order extension (Section 5.4) uses standard quantifiers rather than the N-operator, and notes the finite-domain connection as a separate theorem to be proved only when an N-operator formalization is available.

Proof assistants and philosophy. Benzmüller and colleagues [8] have used Isabelle to formalize ontological arguments and modal logic via shallow semantic embeddings of higher-order modal logic. Fitelson and Zalta [9] pioneer "computational metaphysics," using automated reasoners to verify theorems in abstract-object theory. Scherf [10] provides the most directly comparable work: a machine-checked axiomatization of Advaita Vedānta in Lean 4 (69 axioms, 40+ theorems), demonstrating that complete philosophical systems can be formalized in dependent type theory. Corfield [11] argues at book length that homotopy type theory is a better logic for philosophy than predicate logic, but provides no machine-checked formalizations. To our knowledge, no prior work formalizes any part of the *Tractatus* in any proof assistant, nor uses the type system to isolate the saying/showing boundary as a provable constraint.

7 Discussion

7.1 What the Formalization Shows

The formalization yields three kinds of insight.

First, it makes the *Tractatus*'s implicit modeling decisions explicit. The choice $\mathsf{World}\ \mathsf{S} = \mathsf{S} \to \mathsf{Prop}$ is not neutral: it builds independence into the foundations. By parameterizing over `WorldModel`, we separate what follows from the propositional syntax (compositionality) from what follows from the choice of world space (independence). This distinction is available in principle but rarely made precise in the philosophical literature.

Second, it transforms the saying/showing distinction from a metaphilosophical thesis into a theorem with a two-line proof. The `expresses` relation is the key: once the object/meta boundary is a type distinction, the collapse result (Theorem 5.2) follows by transitivity of biconditional. The philosophical content lies not in the proof's difficulty but in the *formulation*: the act of making `expresses` a typed relation reveals a structural constraint that is invisible when the object/meta boundary is handled informally.

Third, the equivalence hierarchy (Theorem 5.5) provides a precise formal candidate for Wittgenstein's notion of "logical form." The strict separation $\sim_{\text{struct}} \subsetneq \sim_{\text{form}} \subsetneq \sim_{\text{sem}}$ shows that logical form, understood as connective structure up to atom permutation, is a genuine intermediate notion that cannot be reduced to either syntax or semantics.

7.2 Limitations

Several aspects of the *Tractatus* lie beyond the scope of this formalization.

Object structure. The type S is abstract: we do not model how objects combine into states of affairs. TLP 2.0141 ("The possibility of its occurrence in atomic facts is the form of the object") suggests a richer dependent-type encoding where the form of an object constrains its combinatorial possibilities. This would make independence a substantive theorem rather than a definitional consequence.

The N-operator. Wittgenstein's claim that all propositions can be generated from elementary propositions by successive applications of a single operation (TLP 6) is not formalized here. The N-operator is a finitary syntactic construction, and its connection to quantification over infinite domains remains philosophically contentious (Section 5.4).

The picture theory. The *Tractatus*'s claim that propositions are *pictures* of reality (TLP 2.1–2.174) involves a notion of structural isomorphism between language and world that goes beyond truth-functional evaluation. Our formalization captures the truth-conditional aspect but not the pictorial-isomorphism aspect.

Self-reference and the ladder. TLP 6.54 famously declares that the propositions of the *Tractatus* itself are "nonsensical" and must be "thrown away" after use. Our formalization is, by design, a metalanguage that talks *about* the Tractarian object language. It cannot formalize its own status as a ladder to be discarded—this is the point where type theory, like any formal system, reaches its own saying/showing boundary.

We mark this boundary with a deliberate axiom:

```
axiom silence : True
```

This is not a missing proof but a design boundary. The axiom states nothing contentful (`True` is already provable); it marks the point where the formalization acknowledges its own limits.

8 Conclusion

We have presented a machine-checked reconstruction of the semantic core of *Tractatus Logico-Philosophicus* in Lean 4. The formalization proves 46 theorems, decomposes the *Tractatus* into structural invariants, model-dependent assumptions, and formal limits of expressibility, and characterizes the saying/showing distinction as a theorem: nontrivial propositions cannot express world-independent truths.

The equivalence hierarchy $\sim_{\text{struct}} \subsetneq \sim_{\text{form}} \subsetneq \sim_{\text{sem}}$ provides a typed decomposition of the space between syntax and semantics, isolating logical form as a genuine intermediate structure.

These results suggest that proof assistants are a productive tool for philosophical analysis—not merely for verifying known results, but for revealing structural constraints that are invisible when the analysis is conducted on paper.

The formalization is publicly available at `https://leangenius.org/proof/tractatus-ontology`.

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